

Michael Zang

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EDUCATION

University of Southern California | Los Angeles, CA

Aug 2026 – May 2028 (Expected)

M.S. Computer Science

University of Notre Dame | Notre Dame, IN

Aug 2022 – May 2026

B.S. Computer Science, Minor in Business Economics

Coursework: Data Science, AI, Database, Algorithms, Data Structures, Linear Algebra, Calculus, and Statistics

TECHNICAL SKILLS

Languages & Analytics: Python (Pandas, NumPy, Scikit-learn), SQL (MySQL, PostgreSQL), R, Excel
Machine Learning & Statistics: Regression, Classification, Clustering, A/B & Hypothesis Testing, Time-Series Analysis
Data Visualization & BI: Tableau, Power BI, Streamlit, Matplotlib, Seaborn
Big Data & Cloud: PySpark, AWS (S3, EC2), Databricks
Tools: Git, Linux, Jupyter, Docker

PROFESSIONAL EXPERIENCE

Data Scientist Intern | Miami, FL

May 2025 – July 2025

Carnival Cruise Line – Fuel Performance & Decarbonization

- Engineered a regression-based fuel consumption model in Python with a custom GUI, reverse-engineering Eniram's proprietary logic. Scoped for fleet-wide deployment across 29 ships to reduce fuel costs and emissions.
- Designed a data pipeline using MMDetection and OCSort to process live CCTV feeds from 50+ cameras with minute-level scheduling. Stored results in Amazon Neptune for graph-based HVAC efficiency analytics.
- Created interactive dashboards to visualize key performance metrics, enabling stakeholders to monitor fuel usage patterns and identify operational inefficiencies in real time.

Data Scientist Intern | Beijing, China

June 2024 – Aug 2024

Lenovo Group – Lenovo Research AI Lab

- Applied association rule mining (Apriori), decision trees, and LLMs to uncover hidden stock price relationships among major competitors, achieving a 79.42% confidence rate between Lenovo, SMIC, Digital China, and Sunny.
- Queried and processed a 30-million-row SQL database, integrating NLP sentiment signals with traditional ML models into a unified stock prediction model, achieving R^2 scores up to 0.81.
- Presented model performance, feature insights, and potential trading strategies to senior researchers.

Computer Vision Research Assistant | Notre Dame, IN

Aug 2024 – Aug 2026

University of Notre Dame – CSE | First Author (arXiv, WACV 2027)

- Curated a 2,000-image “in-the-wild” monozygotic twin dataset to investigate facial recognition model behavior.
- Evaluated state-of-the-art face recognition models (ArcFace, AdaFace, UniFace, MagFace) using 10-fold cross-validation, achieving 76.50% verification accuracy.
- Retrained ArcFace without flip augmentation and applied RISE saliency analysis to identify key distinguishing features (e.g., mirror twins, skin marks), providing insights into biometric security and model fairness.

SELECTED PROJECTS

Steam Game Lifecycle Analysis | Full-stack | [GitHub](#)

Aug 2026 – Present

- Conducting large-scale analysis on 600K+ monthly records across 5,700+ Steam games using K-Means clustering and hypothesis testing (t-tests, ANOVA) to identify lifecycle patterns and success factors of game longevity.
- Trained a Random Forest classifier (63% accuracy) to predict lifecycle stages, with an interactive Streamlit dashboard to visualize trends and share insights with users.

Closet Recommendation Website | Project Lead | [GitHub](#)

Sep 2024 – Dec 2026

- Led a team researching and implementing recommender algorithms in PyTorch, including Matrix Factorization, Neural Collaborative Filtering (NCF), and LightGCN.
- Cleaned and structured a 30,000-item dataset using a multi-stage regex pipeline and analyzed user-item interaction patterns through EDA. Built an outfit recommendation backend with MySQL and JWT authentication.

LEADERSHIP AND ACTIVITIES

Student International Business Club (Amazon) | Notre Dame, IN

Feb 2022 – May 2024

- Designed a chatbot for Amazon Health, addressing mental health challenges by connecting users directly with professionals, reducing fees, and fostering trust. Presented to executives in the Boston office.

CS For Good | Notre Dame, IN

Sep 2023 – May 2024

- Led the Dismas House project, focused on the reintegration of formerly incarcerated individuals into society.
- Modernized the organization's website, updated the resident ID database, and conducted technology classes.